



PicoMagnetometer User manual

<https://ukraa.com/>

The UK Radio Astronomy Association

A charitable incorporated organisation

Registered charity in England and Wales No. 1123866

Pico^{nT}



Acknowledgements

Design Team

The UKRAA PicoMagnetometer was developed by Paul Hearn, Richard Knott and Andrew Thomas from the original work of the Cosmic Watch detector.

Testing Team

The UKRAA PicoMagnetometer was tested by Paul Hearn, Andrew Thomas, Richard Knott and John Thain.

Production Team

The initial batch of the UKRAA PicoMagnetometer was produced by Paul Hearn and Richard Knott.

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Introduction

UKRAA

The UK Radio Astronomy Association (UKRAA) is a non-profit-making charitable incorporated organisation.

It was established by the Radio Astronomy Section of the British Astronomical Association (BAA) to facilitate the production and sale of radio astronomy products.

This Manual describes how to connect and use the UKRAA PicoMagnetometer detector.

Any suggestions or recommendations for improvement of this Manual would be appreciated. See the **Contacts** page for further details.

The UKRAA PicoMagnetometer



The UKRAA PicoMagnetometer and sensor head

The PicoMagnetometer detector will detect changes in the Earth's magnetic field

To eliminate the recording of β -particles and γ -rays from cosmic ray generated muons, the UKRAA PicoMagnetometer detector has two scintillator assemblies, separated by 17.0mm, that can directly record coincidence events associated with cosmic ray muons.

The detected events are displayed on the OLED screen, output through a serial connection or recorded to an inserted microSD card.

The PicoMagnetometer detector is an active device and requires a +5V supply via a micro-USB cable from suitable wall outlet, power-bank or suitable computer with USB type A socket.

Support

All users of the UKRAA PicoMagnetometer are encouraged to make use of the support available from UKRAA, and others, for setting up and operation the magnetometer.

For ongoing support there is a user forum where discussion, questions and ideas can be shared.

Please see the **Contacts** section for details.

Unboxing and first power on

You will receive your PicoMagnetometer in a white postal box.



Upon opening the box you will see the included micro-USB cable, for connecting your PicoMagnetometer to a suitable +5V DC supply – this can be a USB power outlet, a USB connector on a PC or a USB power bank.

Along with supplied microSD card – this contains your detectors test results, manuals and software.

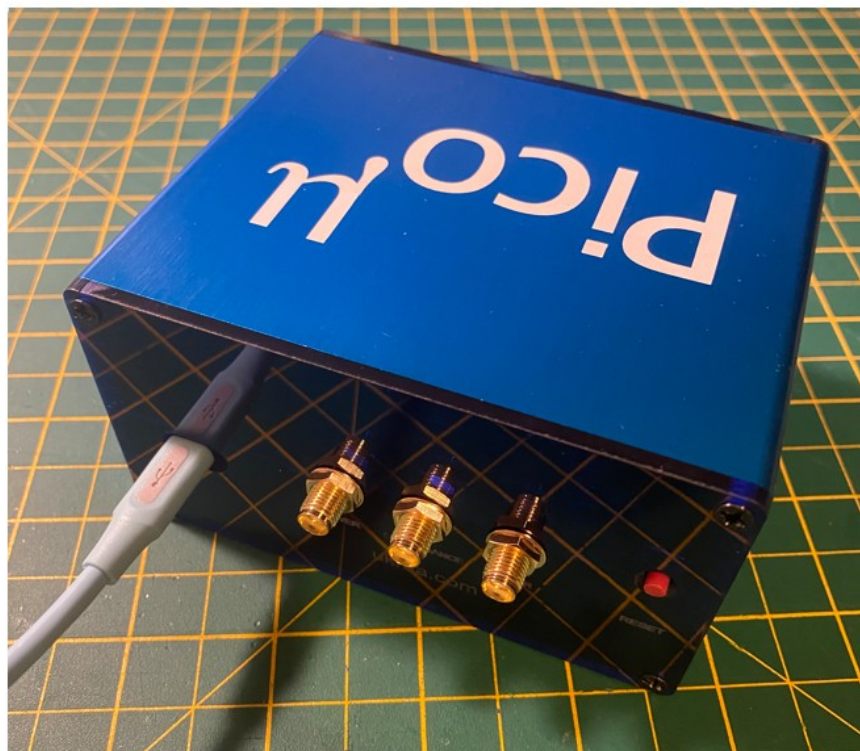


Removing the micro-USB cable, the PicoMagnetometer is wrapped in corrugated cardboard for transit. Remove the cardboard and unwrap to reveal the PicoMagnetometer

NOTE: Please retain the corrugated cardboard and the white postal box should you have need to return the unit to UKRAA.



Plug the micro-USB cable into the micro-USB socket located from the rear panel of the PicoMagnetometer.



Plug the USB type A connector into a suitable USB +5V DC adapter.

Note: picture indicates UK power plug – other countries power plugs will be different.



Insert the USB +5V DC adapter into a mains outlet and switch on.

Note: picture indicates UK power socket – other countries power sockets will be different.



The PicoMagnetometer draws less than 1W, as measured with a USB power meter.

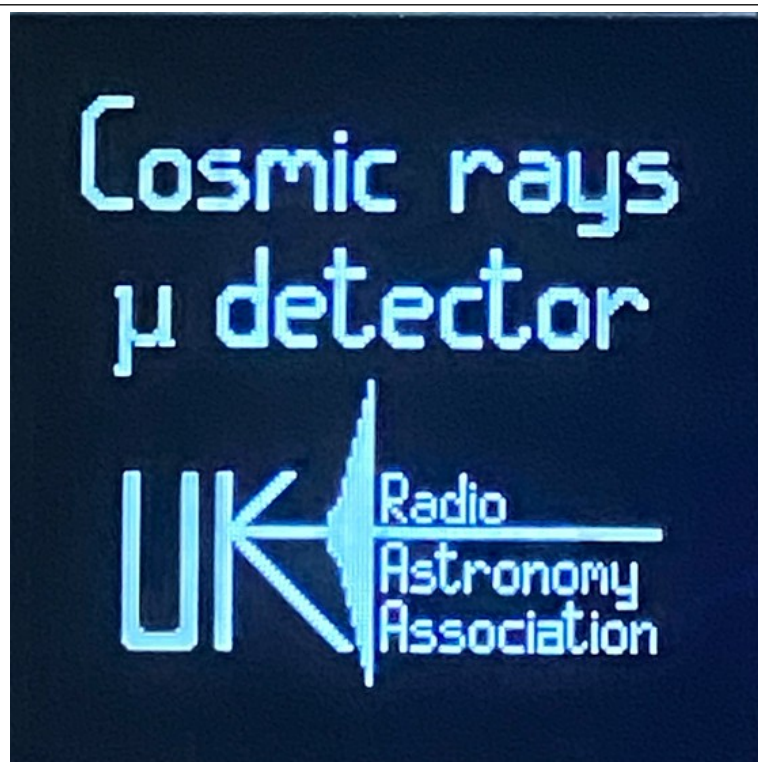
So any suitable USB type A +5V DC adapter that can supply more than 1W should be suitable for powering the unit.



The PicoMagnetometer is now powered.

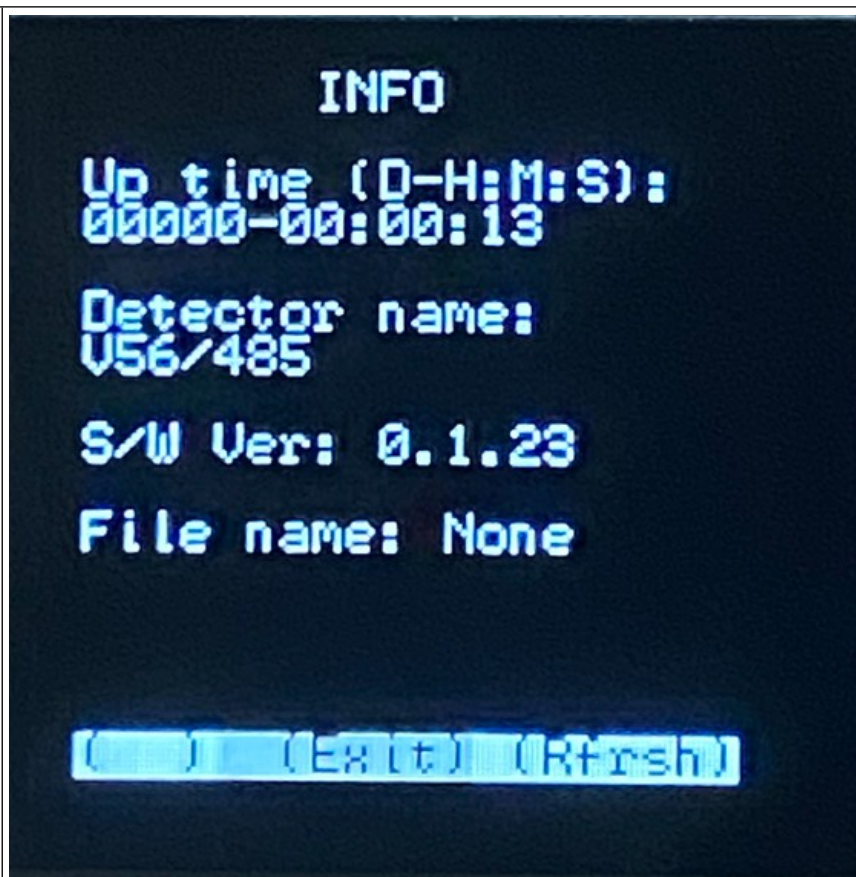
Observing the front panel you will see all three LEDs illuminated at 100% for 5 seconds.

You will then see all three LEDs illuminated at preset user configurable brightness as well as the UKRAA PicoMagnetometer splash screen for 5 seconds.



You will now see the **INFO** screen displayed, this is described in the **OLED display – INFO screen** – see this section for more information.

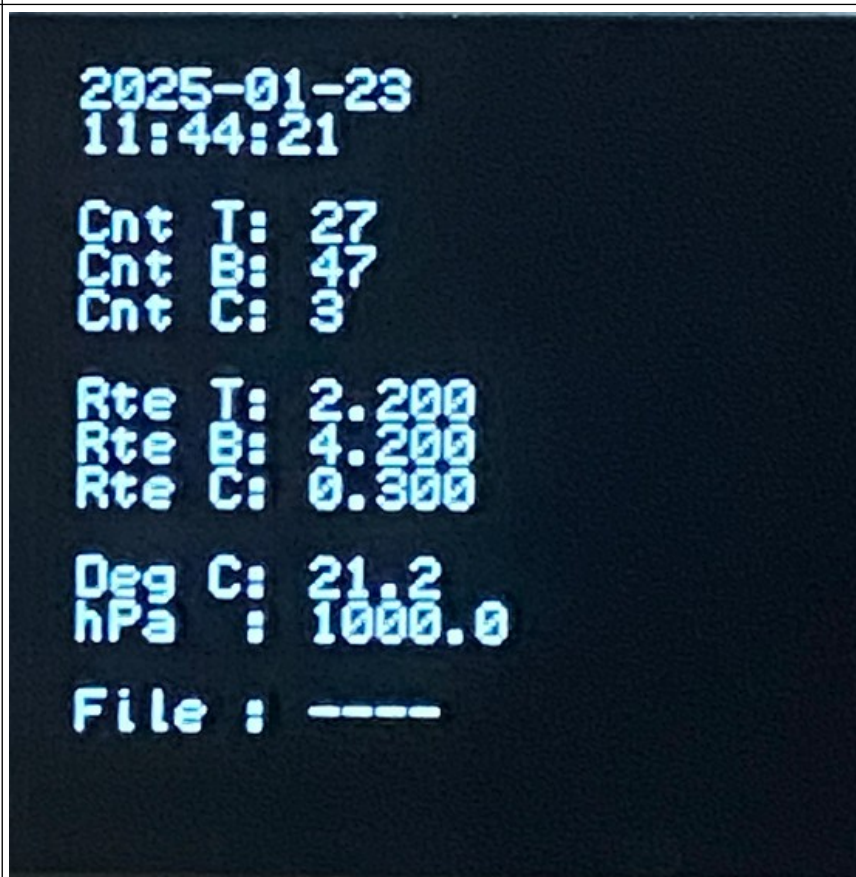
The INFO screen will be displayed for 3 second.



You will now see the **Runtime Display** screen displayed, this is described in the **OLED display – Runtime Display** – see this section for more information.

Congratulations – your PicoMagnetometer is now detecting β -particles, γ -rays and muons.

The PicoMagnetometer will remain on this screen, unless user input from the button switches located on the front panel.



Front panel

DOWN button

Used to control the cursor on the OLED display screen when moving between menus

ENTER button

Used to access the PicoMuon menus and confirm action

UP button

Used to control cursor on the OLED display screen when moving between menus



OLED display

128x128 pixel monochrome display.

Displays various information relating to the function / working of the PicoMuon display.

MicroSD socket

Used to input user configuration parameter variables, via detector.ini file, on microSD card

Used to store event data on microSD card

Rear panel

RPi Pico micro-USB

Plug suitable micro-USB cable into this port to power the PicoMuon detector and enable data transfer from the PicoMuon detector to suitable serial interface

Reset switch

Press to action

Will restart the PicoMuon detector when pressed.

Also used to enable upload of new UF2 code in conjunction with RPi Pico **BOOT SEL** button.



Lower event sma connector

Lower detector event pulse can be accessed via this sma connector.

See Typical scope traces for shape of pulse

Coincidence event sma connector

Coincidence event pulse can be accessed via this sma connector.

See Typical scope traces for shape of pulse

Upper event sma connector

Upper detector event pulse can be accessed via this sma connector.

See Typical scope traces for shape of pulse

OLED display – Runtime Display screen

Date and time

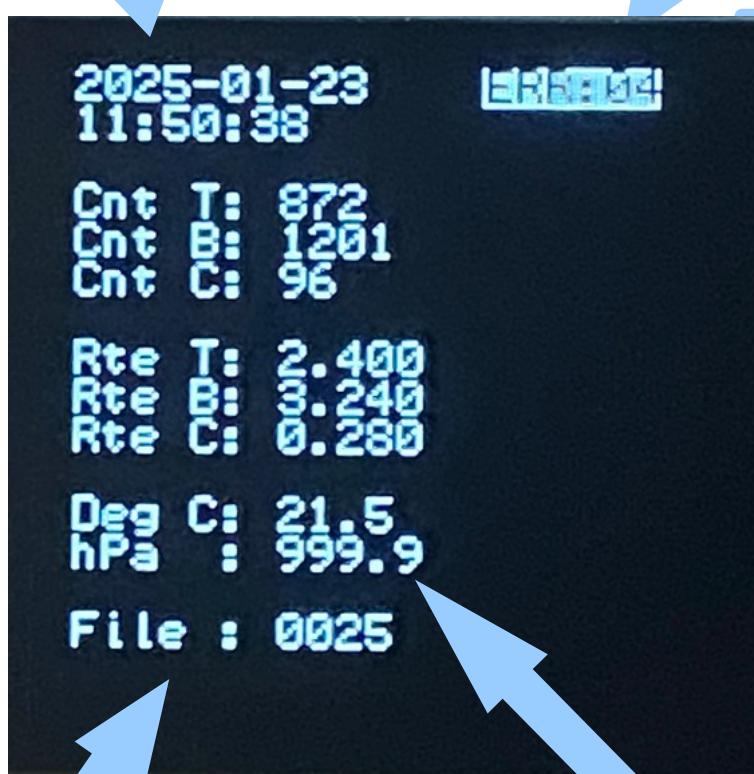
The current date (YYYY-MM-DD) and time (HH:MM:SS), taken from the DS3231 RTC at start-up, will be displayed here.

The time will increment each second.

ERROR code

Any error code will be displayed in the top right of the OLED display

See ERROR codes section for meaning



Counts

The total counts for the top (T), bottom (B) and coincidence (C) events are recorded here.

Count rates

The current count rate for the top (T), bottom (B) and coincidence (C) events are recorded here.

This is a rolling count rate for the last 60 seconds – in units of counts per second

File

Provides the current file name that is being used for saving event data to when a microSD card is inserted into the microSD card slot.

If no microSD card inserted will display

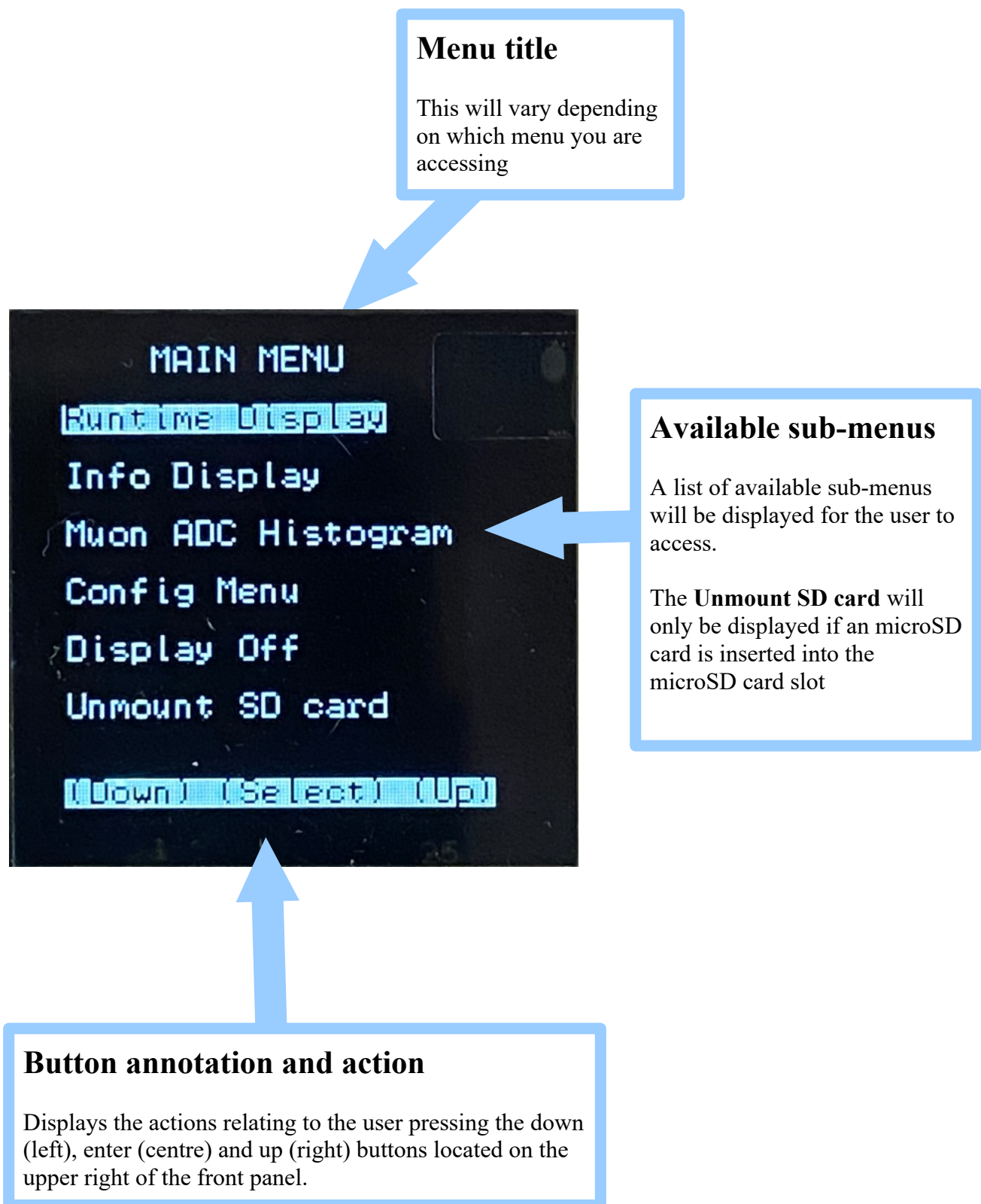
File : ----

Temperature and pressure

Displays the current temperature and pressure recorded from the BMP280 sensor.

Temperature is in deg C and pressure is in hPa

OLED display – MAIN MENU screen



OLED display – INFO screen

The **Info Display** is accessed through the **MAIN MENU** by using the **DOWN** button to highlight **Info Display** and then pressing the **ENTER** button.

Menu title

This will vary depending on which menu you are accessing

Detector uptime

This displays the time, in days-hours:minutes:seconds, since the detector has been powered on.

Selecting refresh will refresh the uptime displayed.

Detector name

User defined detector name.

Default is model-serial number.

Changed by user using detector.ini file.

Software Version

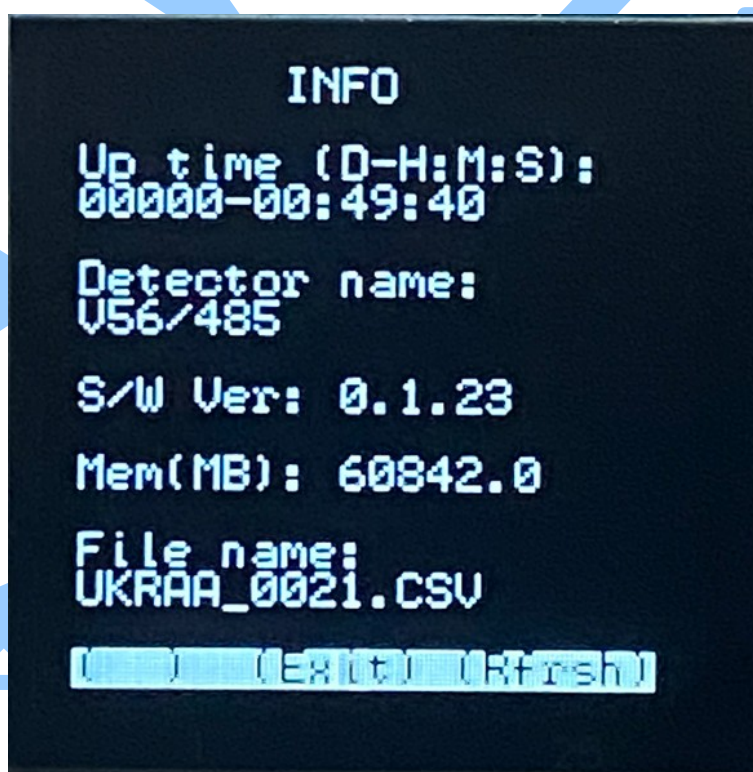
Current version of the running software on the PicoMuon detector.

As bugs/enhancements are identified – the core code will need to be updated by the user. Follow **Uploading updated software** for instruction.

File name

Full file name (including user defined file prefix – here **UKRAA_**) of file used to store event data on inserted microSD card.

If no microSD card inserted will display **File name: None**



Available memory on microSD

Current free capacity on the inserted microSD card.

As event data is stored to the microSD card – this will decrease.

If no microSD card is inserted then this line is not displayed..

Button annotation and action

Displays the actions relating to the user pressing the down (left), enter (centre) and up (right) buttons located on the upper right of the front panel.

OLED display – Muon ADC Histogram screen

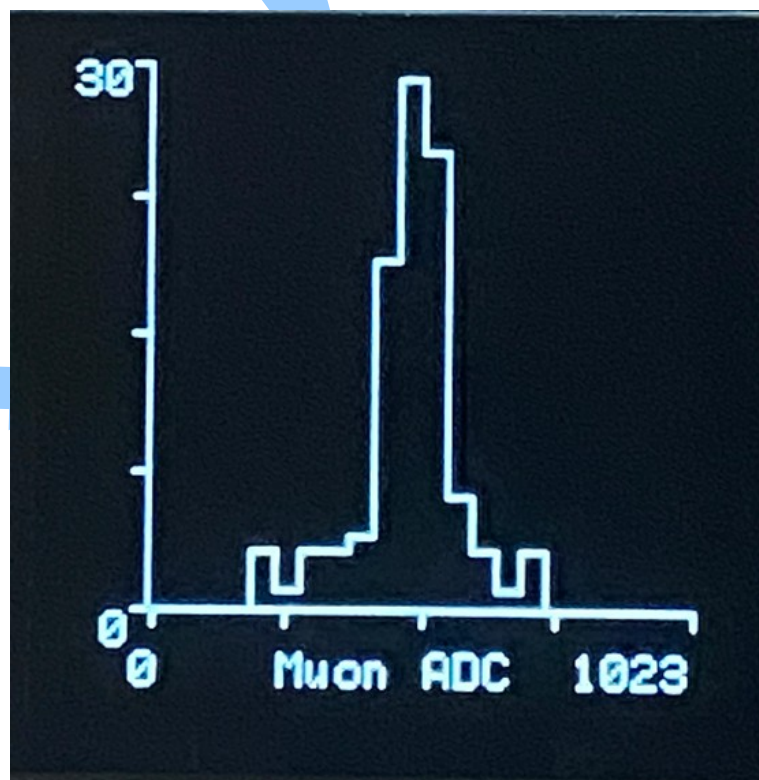
ADC histogram

The graph will update as time progresses and provides a rolling 5 minute snap shot of the recorded ADC values associated with coincidence (muon) events recorded by the PicoMuon detector.

Frequency of ADC value

Y-axis represents the frequency of ADC values recorded for the last 100 events.

Scale is 0 to 30.



Muon ADC value

X-axis represents the ADC values of recorded muon events in the last 5 minutes.

Scale is 0 to 1023 ADC units, binned at 20 ADC units – remember there is a threshold below which event data is not recorded.

Although the PicoMuon detector is not calibrated – the recorded ADC values gives an indication of the energy deposited by the ionising radiation event recorded by the detector

OLED display – CONFIG MENU screen

Browse Config

Enables user to scroll through the current EEPROM stored configuration parameter variables

Load Config

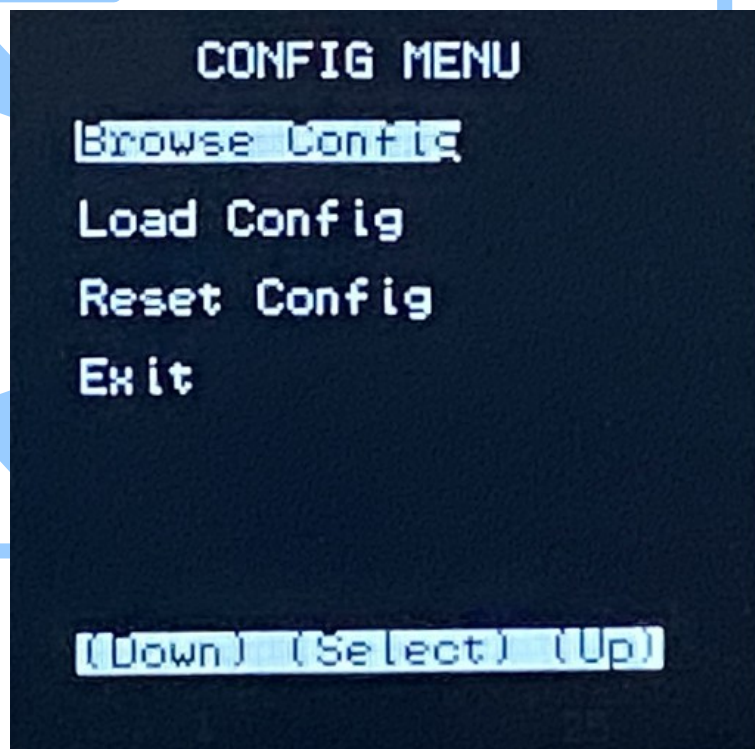
Enables user to upload new configuration parameter variables to the EEPROM

Reset Config

Enables user to reset the user configurable parameter variables stored in the EEPROM

Exit

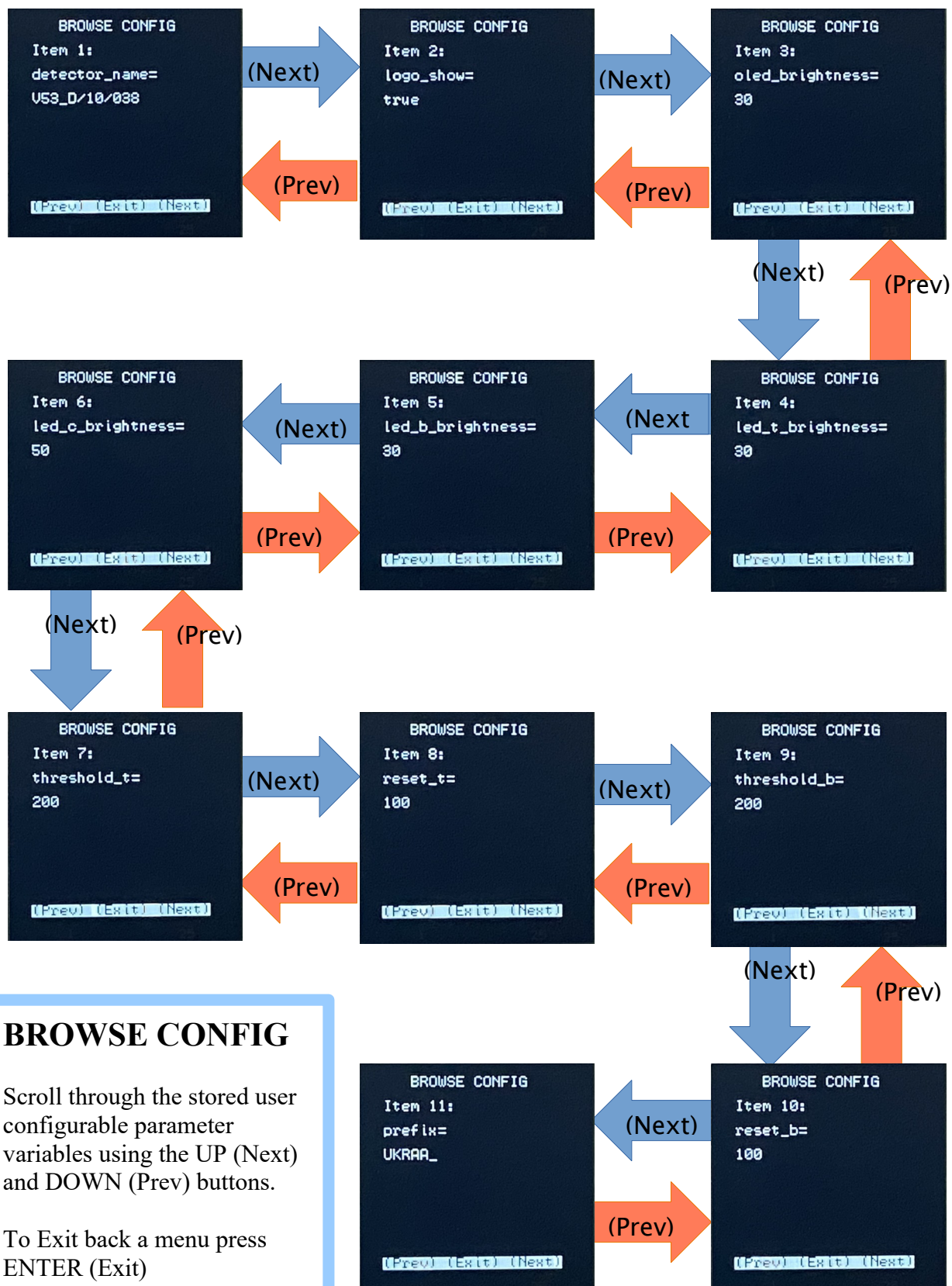
Enables user to exit this menu.



Button annotation and action

Displays the actions relating to the user pressing the down (left), enter (centre) and up (right) buttons located on the upper right of the front panel.

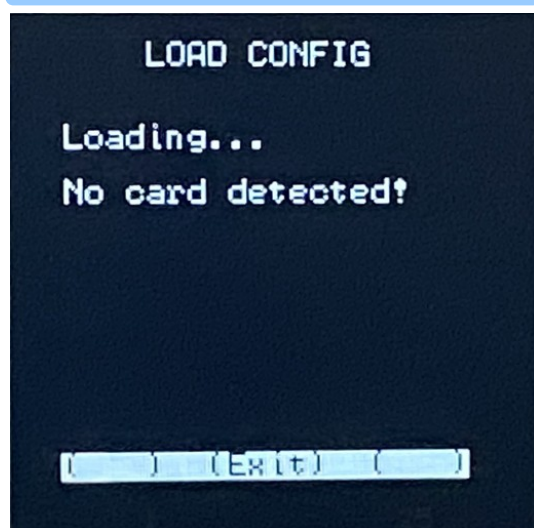
OLED display – BROWSE CONFIG screens



OLED display – LOAD CONFIG screens

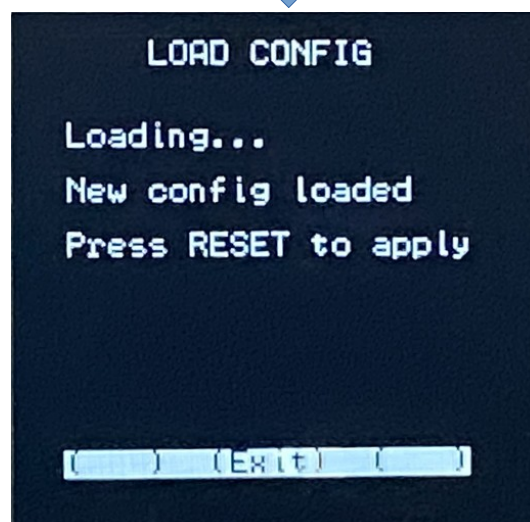
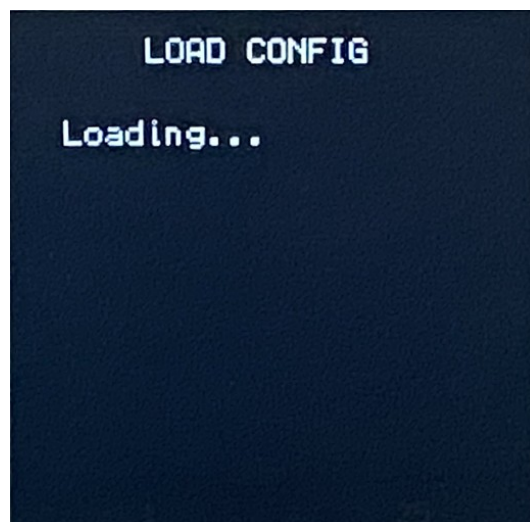
Selecting the **Load Config** option from the **CONFIG MENU** screen will bring up one of two screens

No microSD card inserted into the microSD slot



Action:
Press ENTER (Exit)
Insert microSD card into microSD card slot
Re-select Load Config

microSD card inserted into the microSD slot – with valid detector.ini file



The new configuration has been stored to the EEPROM.

The changed user parameter variables will be used once the PicoMuon detector is RESET.

Press the **RESET** switch located on the rear panel to apply new configuration variables

OLED display – Invalid LOAD CONFIG screen

If there was any deviation from the expected user configurable parameter variables within the **detector.ini** file on the microSD card when loading new user configurable parameter variables – eg the inclusion of an alpha character when expecting a numerical character, or the variable is outside of the expected numerical range for that parameter, then a **Failed!** message will be displayed, along with the reason.

Invalid character

```
[display]
oled_brightness = 3e0
```

Action: correct error in **detector.ini** file and redo.

LOAD CONFIG

```
Loading...
Failed!
oled_brightness=
Invalid character
```

```
( ) (Exit) ( )
```

Invalid value - 1

```
[leds]
led_t_brightness = 300
```

Action: correct error in **detector.ini** file and redo.

LOAD CONFIG

```
Loading...
Failed!
led_t_brightness=
Must be (0-100)
```

```
( ) (Exit) ( )
```

Invalid value - 2

```
[event]
threshold_t = 200
reset_t = 300
```

Action: correct error in **detector.ini** file and redo.

LOAD CONFIG

```
Loading...
Failed!
reset_t must be
< threshold_t
```

```
( ) (Exit) ( )
```

OLED display – RESET CONFIG screen

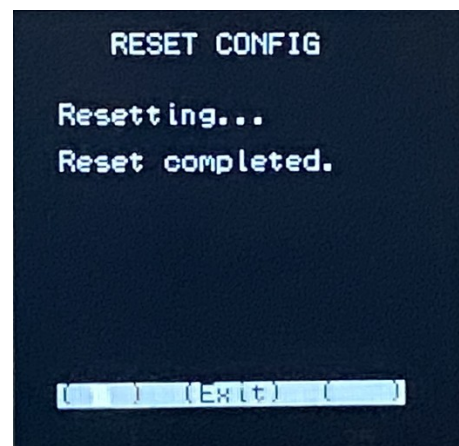
Selecting the **Reset Config** option from the **CONFIG MENU** screen will bring up the following screen

This will reset the stored user configurable parameter variables,



Once complete you will see the following screen.

Press **ENTER** (Exit) to return to previous menu screen.



Base configuration values after RESET CONFIG actioned

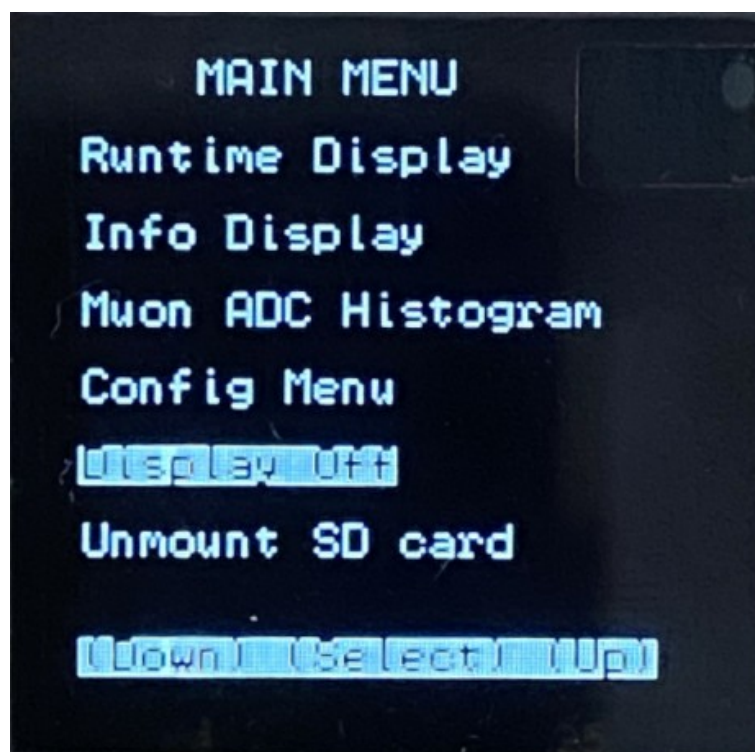
```
detector_name=Undefined
logo_show=true
oled_brightness=30
led_t_brightness=30
led_b_brightness=30
led_c_brightness=50
threshold_t=200
reset_t=100
threshold_b=200
reset_b=100
prefix=UKRAA_
```

OLED display – Display off

OLED displays have a unique feature of “burn in” – where if the display has the same unchanging text for a long period of time then some of the pixels will either fade or not function.

To avoid this happening you have the opportunity to turn your OLED display off when not needed.

This is available through the **MAIN MENU**.



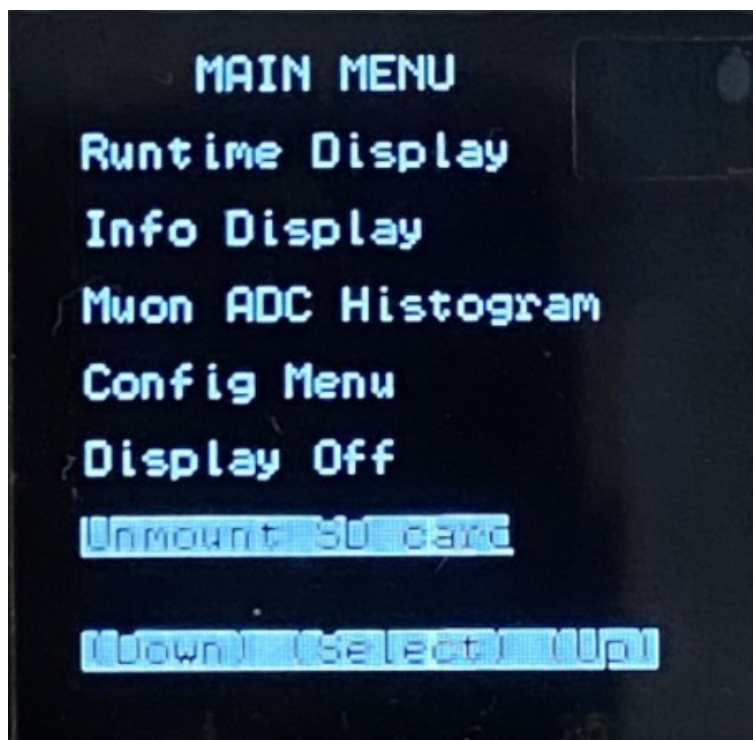
Selecting this option when the OLED display is not needed will improve the longevity of the PicoMuon OLED display.

Pressing the **ENTER** button on the front panel will bring the OLED display back on.

OLED display – Unmount SD card

To be able to safely eject an inserted microSD card, there is an option available via the **MAIN MENU** that enables this to happen.

It is always sensible to **Unmount** the inserted microSD card from the PicoMuon detector – rather than just pulling it out while still writing data to it.



Changing user configuration values

- 1) The PicoMagnetometer has user configurable parameter variables, that can be set using a text file named **magnetometer.ini**.
- 2) The structure of the **magnetometer.ini** file is as follows:

```
;;UKRAA PicoNanoTesla Magnetometer--User configuration fileCR1F
;CR1F
; Follow the instructions in the UKRAA PicoNanoTesla Magnetometer manual for editing this file CR1F
;CR1F
; Follow the instructions in the UKRAA PicoNanoTesla Magnetometer manual for using this fileCR1F
;CR1F
CR1F
[Name].....; detector name configurationCR1F
detector_name = V8/001/KNO.....; 16 characters maximum (EEPROM page 01) CR1F
CR1F
[Start up].....; Show logo/info screen on OLED at start-upCR1F
SlowStart = true.....; [true] = include info screen, false = don't show info screen (EEPROM page 02) CR1F
CR1F
[display].....; Display configurationCR1F
display = true.....; [true] = use OLED display, false = don't use OLED display (EEPROM page 03) CR1F
display_brightness = 30.....; Brightness of OLED display -- [0 to 100]% (EEPROM page 04) CR1F
CR1F
[led].....; LED configurationCR1F
LED = true.....; [true] = use LED display, false = don't use LED displayCR1F
led_brightness = 30.....; Brightness of LED -- [0 to 100]% (EEPROM page 05) CR1F
CR1F
[event].....; Event configuration (EEPROM page 06) CR1F
threshold_1 = 50.....; 1st threshold level (EEPROM page 07) CR1F
threshold_2 = 100.....; 2nd threshold level (EEPROM page 08) CR1F
threshold_3 = 200.....; 3rd threshold level (EEPROM page 09) CR1F
CR1F
[microsd].....; microSD configurationCR1F
prefix = UKRAA_.....; microSD card filename prefix (EEPROM page 10) CR1F
CR1F
[LAN].....; LAN configurationCR1F
LAN_IP = [192, 168, 1, 70].....; static IP address (EEPROM page 11) CR1F
LAN_MASK = [255, 255, 255, 0].....; MASK from your home router (EEPROM page 12) CR1F
LAN_GATEWAY = [192, 168, 1, 254].....; GATEWAY from your home router (EEPROM page 13) CR1F
CR1F
[NTP_Server].....; NTP Server configurationCR1F
NTP_IP_1 = [139, 143, 5, 30].....; NPL NTP Server 1 (EEPROM page 14) CR1F
NTP_IP_2 = [139, 143, 5, 31].....; NPL NTP Server 2 (EEPROM page 15) CR1F
CR1F
[GPS].....; GPS configurationCR1F
GPS_Min_Satellites = 3.....; minimum satellites required for valid GPS fix (EEPROM page 16) CR1F
CR1F
[Environment].....; Environment configurationCR1F
min_tmp36_temp = -10.....; minimum expected temperature for sensor head (EEPROM page 17) CR1F
max_tmp36_temp = 100.....; maximum expected temperature for sensor head (EEPROM page 18)
```

- 3) The **magnetometer.ini** file is separated into 10 sections, that the user can modify to their own requirements.
- 4) If you wish to edit the **magnetometer.ini** file, it is recommended to use a plain text editor such as “notepad++”, <https://notepad-plus-plus.org/>.
- 5) Only change the text after the “=” for each of the parameters, Do not change the parameter name – these are used by the PicoMagnetometer code to register and store the correct variable to the onboard EEPROM.
- 6) If **magnetometer.ini** is modified – save the modified file and then copy onto a suitable microSD card – do not change the file name.
- 7) Insert this microSD card into the microSD card slot located on the front panel of the PicoMagnetometer.
- 8) Access the menu screen of the PicoMagnetometer by pressing the rotary encoder located on the front panel of the PicoMagnetometer. Navigate to the **Config Menu** and then select **Load Config**.

- 9) Once the parameters and variables from the **magnetometer.ini** file have been stored to the onboard EEPROM of the PicoMagnetometer , removal of power will not remove/alter the stored parameter variable values.

Table of parameters/variables from detector.ini file

Parameter	Variable
detector_name	Name the user wishes to describe their PicoMagnetometer Can be maximum of 16 characters Preset value = detector model~serial number – e.g. V1~002
SlowStart	Show UKRAA logo at start up “true” = show logo – this is the preset value “false” = don’t show logo Preset value = true
display	OLED display on/off “true” = OLED display on “false” = OLED display off Preset value = true
display_brighness	Sets the brightness of the OLED display, set between 0 to 100 0 would be off. Preset value = 30
LED	LED on/off “true” = LED on “false” = LED off Preset value = true
led_brightness	Sets the brightness of the LED, set between 0 to 100 0 would be off. Preset value = 30

Parameter	Variable
threshold_1	Sets the lower threshold for the activity plot Value represents nT on the activity plot Must be lower than threshold_2 Preset value = 50
threshold_2	Sets the middle threshold for the activity plot Value represents nT on the activity plot Must be greater than threshold_1 and lower then threshold_3 Preset value = 100
threshold_3	Sets the upper threshold for the activity plot Value represents nT on the activity plot Must be greater than threshold_3 Preset value = 200
prefix	Sets the prefix of the filename when storing data to microSD. Use ASCII text characters. Preset value = UKRAA_
LAN_IP	Sets static LAN IP address for the PicoMagnetometer The PicoMagnetometer should default to DHCP, therefore this should not be required Preset value = 192.168.0.70
LAN_MASK	Sets LAN mask IP address for the PicoMagnetometer This should be changed to the mask IP address for the users ISP router Preset value = 255,255,255,0
LAN_GATEWAY	Sets LAN gateway IP address for the PicoMagnetometer This should be changed to the gateway IP address for the users ISP router Preset value = 192.168.1.254
NTP_IP_1	Sets IP address NTP server These NTP IP addresses are for the UK NTP server; users in different territories will have to source alternative NTP server IP addresses

threshold_2	Sets the middle threshold for the activity plot Value represents nT on the activity plot Must be greater than threshold_1 and lower then threshold_3 Preset value = 100
	Preset value = 139,143,5,30
NTP_IP_2	Sets IP address NTP server These NTP IP addresses are for the UK NTP server; users in different territories will have to source alternative NTP server IP addresses Preset value = 139,143,5,31

Parameter	Variable
GPS_Min_Sattelites	Sets the minimum number of GPS satellites in use for valid date/timestamp The more satellites used the greater the accuracy of the date/timestamp Preset value = 3
min_tem36_temp	Sets minimum temperature anticipated for the sensor head TMP36 temperature range -40°C to +125°C Preset value = -10.0
max_tmp36_temp	Sets maximum temperature anticipated for the sensor head TMP36 temperature range -40°C to +125°C Preset value = 100

Event data – microSD

- 1) Events detected by the PicoMagnetometer can be stored directly to a suitable microSD card, inserted into the microSD card slot located via the front panel.
- 2) When a microSD card is inserted into the PicoMagnetometer, the card will be detected and a file will be created for storing event data.
- 3) The file name consists of {prefix}+{number}+.csv – where:
 - {prefix} = user *prefix* parameter variable defined in **detector.ini**
 - {number} = sequential number generated by code – starting 000, and incrementing each time PicoMagnetometer is reset, or microSD card is unmounted and then reinserted.
 - .csv = file type defined
 - So typical file name would be UKRAA_0001.csv by default, then incrementing to UKRAA_0002.csv when PicoMagnetometer detector is reset or microSD card is unmounted and then reinserted
- 4) Format of data stored to microSD is as follows:

- Header section:

```
#####
### UKRAA: The Pico Muon Detector                                     ###
### S/W version: 0.1.23                                             ###
### Detector name: V56/485                                          ###
### Thresholds: T(200 ,100 ), B(200 ,100 )                          ###
#####
```

- This is 71 characters wide and 6 lines of plain text, comprising
 - Software version currently being run by the PicoMagnetometer detector
 - The name of the detector
 - The threshold and reset ADC values, set for the upper (T) and lower (B) SiPM assemblies
- There is then a single line that identifies the parameters that are recorded:

Date,Time,ID,Event,ADC(0–1023),ElapsedTime(mS),DeadTime(mS),Temperature(C),Pressure(hPa),DetectorName

- This is 103 characters wide and a single line of plain text, comprising of data column identifiers.
- ID indicates the detector position: T = top (upper), B = bottom (lower) and C = coincidence (muon).

- Finally there is the event data recorded to the microSD card relating to the column headers shown above.

```
2025-01-24,10:21:04,B,1,251,87,1,20.2,992.4,V56/485
2025-01-24,10:21:04,T,1,82,235,0,20.2,992.4,V56/485
2025-01-24,10:21:05,T,2,231,1075,1,20.2,992.4,V56/485
2025-01-24,10:21:05,B,2,92,1193,6,20.2,992.4,V56/485
```

5) Here is an example of a data file opened in notepad:

```
#####
### UKRAA: The Pico Muon Detector          ###
### S/W version: 0.1.23                    ###
### Detector name: V56/485                 ###
### Thresholds: T(200 ,100 ), B(200 ,100 ) ###
#####
Date,Time,ID,Event,ADC(0-1023),ElapsedTime(mS),DeadTime(mS),Temperature(C),Pressure(hPa), DetectorName
2025-01-24,10:21:04,B,1,251,87,1,20.2,992.4,V56/485
2025-01-24,10:21:04,T,1,82,235,0,20.2,992.4,V56/485
2025-01-24,10:21:05,T,2,231,1075,1,20.2,992.4,V56/485|
2025-01-24,10:21:05,B,2,92,1193,6,20.2,992.4,V56/485
2025-01-24,10:21:05,B,3,275,1346,6,20.2,992.4,V56/485
2025-01-24,10:21:05,B,4,94,1398,6,20.2,992.4,V56/485
2025-01-24,10:21:05,B,5,120,1415,6,20.2,992.4,V56/485
2025-01-24,10:21:05,T,3,138,1473,1,20.2,992.4,V56/485
2025-01-24,10:21:05,B,6,603,1632,6,20.2,992.4,V56/485
2025-01-24,10:21:05,B,7,112,2045,6,20.2,992.4,V56/485
```

6) The csv file can be opened in any data processing software for the user to manipulate as necessary. Here is the same data file open in a spreadsheet:

	A	B	C	D	E	F	G	H	I	J
1	#####									
2	### UKRAA: The Pico Muon Detector					###				
3	### S/W version: 0.1.23				###					
4	### Detector name: V56/485				###					
5	### Thresholds: T(200 100)		B(200 100)		###					
6	#####									
7	Date	Time	ID	Event	ADC(0-1023)	ElapsedTime(mS)	DeadTime(mS)	Temperature(C)	Pressure(hPa)	DetectorName
8	24/01/2025	10:21:04	B	1	251	87	1	20.2	992.4	V56/485
9	24/01/2025	10:21:04	T	1	82	235	0	20.2	992.4	V56/485
10	24/01/2025	10:21:05	T	2	231	1075	1	20.2	992.4	V56/485
11	24/01/2025	10:21:05	B	2	92	1193	6	20.2	992.4	V56/485
12	24/01/2025	10:21:05	B	3	275	1346	6	20.2	992.4	V56/485
13	24/01/2025	10:21:05	B	4	94	1398	6	20.2	992.4	V56/485
14	24/01/2025	10:21:05	B	5	120	1415	6	20.2	992.4	V56/485
15	24/01/2025	10:21:05	T	3	138	1473	1	20.2	992.4	V56/485
16	24/01/2025	10:21:05	B	6	603	1632	6	20.2	992.4	V56/485
17	24/01/2025	10:21:05	B	7	112	2045	6	20.2	992.4	V56/485

Event data – serial output

- 1) The PicoMagnetometer provides event data via the serial interface. The format of the data is similar to that recorded to the microSD card, except there is no header and the single line that identifies the parameters that are recorded does not include date or time, as it is anticipated that the computer that the PicoMagnetometer is connected to will provide suitable date and time.

ID,Event,ADC(0–1023),ElapsedTime(mS),DeadTime(mS),Temperature(C),Pressure(hPa),DetectorName

- 2) The user can thus utilise a simple script to record and store the event data from the PicoMagnetometer and then analyse the data as they wish.
- 3) Below is an example of a simple Python script, to record and store the event data from the PicoMagnetometer, run on a RPi4 single board computer.

```
# imports
import serial
from datetime import datetime, timezone

# definitions
def main():
    # set up usb serial variables
    ser = serial.Serial(port = '/dev/ttyACM0',
                        baudrate = 115200,
                        bytesize = serial.EIGHTBITS,
                        parity = serial.PARITY_NONE,
                        stopbits = serial.STOPBITS_ONE)

    for line in ser:
        # define output file to write data to
        outfile = open('/home/pi/UKRAA_PicoMuon/data.txt', 'a')
        # if data, write to file
        if line:
            # date and time
            timetowrite = (datetime.strftime(datetime.now(timezone.utc), '%Y-%m-%d')
                          + " "
                          + datetime.strftime(datetime.now(timezone.utc), '%H:%M:%S.%f'))
            # data received from PicoMuon detector
            texttowrite = (line.decode('utf-8', 'ignore').strip())

            print(timetowrite + ',' + texttowrite, file = outfile)

# main
if __name__ == "__main__":
    main()
```

- 4) The data stored from the above script has the following format:

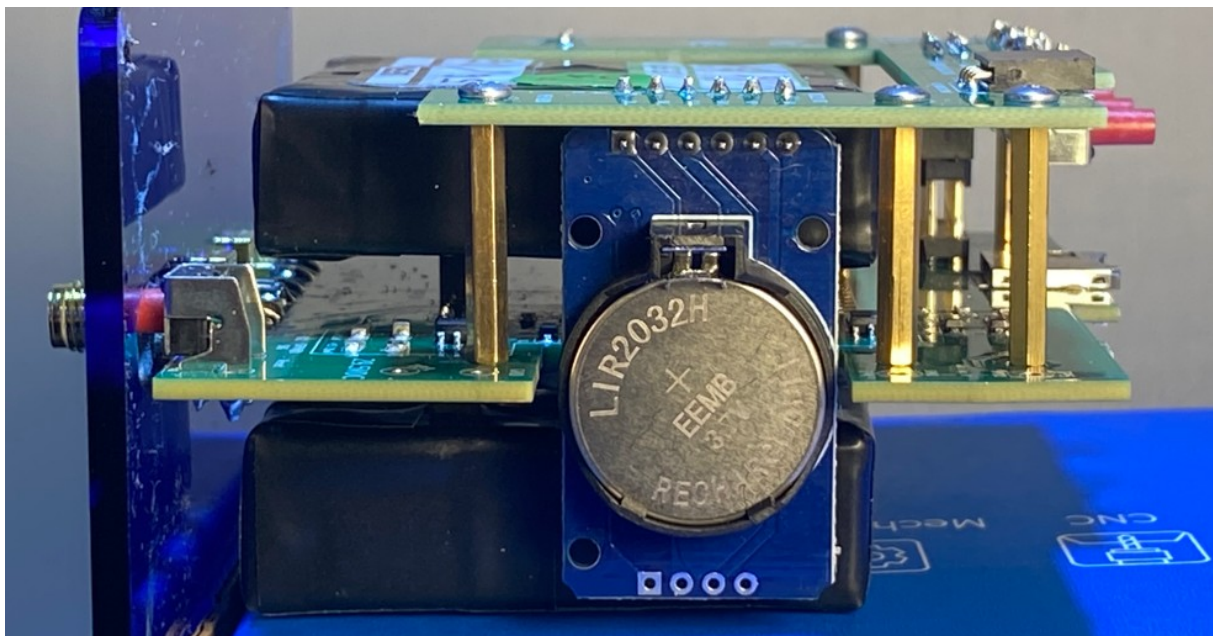
```
2025-01-24 12:16:04.418779,T,15217,54,6901909,8232,20.6,993.5,V56/485
2025-01-24 12:16:04.419031,B,21260,116,6901959,19132,20.6,993.5,V56/485
2025-01-24 12:16:04.419245,B,21261,70,6902080,19132,20.6,993.5,V56/485
2025-01-24 12:16:04.419610,B,21262,51,6902395,19132,20.6,993.5,V56/485
2025-01-24 12:16:05.427845,B,21263,239,6903081,19133,20.6,993.5,V56/485
2025-01-24 12:16:05.428655,T,15218,66,6903615,8232,20.6,993.5,V56/485
2025-01-24 12:16:06.438108,B,21264,60,6903921,19137,20.6,993.5,V56/485
2025-01-24 12:16:06.438560,T,15219,420,6903935,8235,20.6,993.5,V56/485
2025-01-24 12:16:06.438830,T,15220,60,6904168,8235,20.6,993.5,V56/485
2025-01-24 12:16:06.439119,B,21265,276,6904326,19137,20.6,993.5,V56/485
2025-01-24 12:16:06.439362,T,15221,433,6904416,8235,20.6,993.5,V56/485
2025-01-24 12:16:06.439585,B,21266,579,6904518,19137,20.6,993.5,V56/485
2025-01-24 12:16:06.439784,T,15222,444,6904518,8235,20.6,993.5,V56/485
2025-01-24 12:16:06.439990,B,21267,253,6904795,19137,20.6,993.5,V56/485
2025-01-24 12:16:07.447902,T,15223,205,6904929,8236,20.6,993.5,V56/485
2025-01-24 12:16:07.448119,T,15224,100,6905097,8236,20.6,993.5,V56/485
2025-01-24 12:16:07.448314,B,21268,350,6905329,19138,20.6,993.5,V56/485
2025-01-24 12:16:07.448723,T,15225,416,6905410,8236,20.6,993.5,V56/485
2025-01-24 12:16:08.456954,B,21269,297,6905980,19140,20.6,993.5,V56/485
2025-01-24 12:16:08.457178,B,21270,139,6906534,19140,20.6,993.5,V56/485
2025-01-24 12:16:08.457792,B,21271,50,6906921,19140,20.6,993.5,V56/485
2025-01-24 12:16:09.466057,T,15226,115,6907005,8238,20.6,993.5,V56/485
2025-01-24 12:16:09.466957,T,15227,138,6907141,8238,20.6,993.5,V56/485
```

- 5) A full set of Python code is available, from the UKRAA GitHub page, that will get, process and present data from the PicoMagnetometer detector onto your smart phone at home using a RPi4 or RPi5.

https://github.com/UKradioastro/UKRAA_PicoMagnetometer

Replace RTC button cell

- 1) Should you need to remove/replace the button cell from the DS3231 RTC, please contact UKRAA for advice/instruction for resetting the DS3231RTC date/time following button cell replacement.
- 2) Should the CR2032 button cell need to be replaced on the DS3231 RTC module, then the following actions need to be taken.
- 3) Unplug the PicoMagnetometer detector from any external power source.
- 4) Remove all x8 case screws, x4 from the front panel and x4 from the rear panel. It is not necessary to remove the x3 ¼" sma nuts and washers – the rear panel can remain attached to the PicoMagnetometer main PCB.
- 5) Pull the front panel forward to disengage the OLED display from the upper OLED PCB – careful not to bend the pins on the 1x4 header of the OLED display secured to the front panel.
- 6) The upper aluminium case can be removed and the PicoMagnetometer PCB assembly removed from the lower case.
- 7) Set the PicoMagnetometer PCB assembly on a flat surface with the CR2032 button cell visible, as shown in the picture below.



- 8) To remove the CR2032 button cell – pull back on the upper spring clip of the button cell holder. This will release the button cell from the button cell holder, and remove.
- 9) Insert a fresh CR2032 button cell into the button cell holder from the lower black clip, push towards the top of the button cell until it clicks in behind the upper metal spring clip. **Ensure that +ve is facing outwards.**

- 10) The date/time will now need to be reset on the DS3231 RTC – contact UKRAA for instruction on completing this before progressing further – see Contact details at end of manual.
- 11) Replace the PicoMagnetometer PCB assembly into the lower aluminium case and then replace the upper aluminium case
- 12) Align the 1x4 head of the OLED display with the 1x4 socket on the upper OLED PCB and push home, again being careful to correctly engage the 1x4 header with the upper OLED PCB.
- 13) Replace all x8 case screws: x4 from the front panel and x4 from the rear panel
- 14) Plug the PicoMagnetometer detector into suitable external power source.
- 15) Battery replaced, real time clock date/time updated – good to go.

Install new release of software

- 1) Like any provided software, there will be certain bugs that were not captured during pre-release and initial-release testing – these are free!
- 2) Once software bug issues have been raised through the GitHub repository for this product <https://github.com/Ukradioastro/CPicoMagnetometer>, and addressed, a new release of software will be issued. This will be in the form of a .UF2 file. All users for whom UKRAA has suitable contact details, will be made aware of this new software release. The user is then able to update their software, should they wish to do so.
- 3) Other software issues that are raised through the GitHub repository for this product <https://github.com/Ukradioastro/CPicoMagnetometer>, and are identified as enhancements may not be expedited into the next release of code – but will be reviewed and included if deemed appropriate.
- 4) To install new release of software, the user has to have access to the **BOOT SEL** button, located on the RPi Pico seen in the upper left of the picture below.



- 5) To gain access to the RPi Pico **BOOT SEL** button, remove the x2 front pane lower aluminium case screws and the x2 rear panel lower aluminium case screws. Then remove the lower aluminium case from the PicoMagnetometer detector.
- 6) On a suitable PC, download the latest release of software for the PicoMagnetometer detector.
- 7) Connect the PicoMagnetometer detector to this PC using the supplied USB cable with the PicoMagnetometer detector.
- 8) While holding the RPi Pico **BOOT SEL** button down also press the **reset** button located on the rear panel of the PicoMagnetometer detector.
- 9) Now release the **Reset** button, while still holding the RPi Pico **BOOT SEL** button down.
- 10) Now release the RPi Pico **BOOT SEL** button.
- 11) You will now be able to access the RPi Pico as a mass storage device on your PC.
- 12) Open a file explorer of the RPi Pico
- 13) Open a file explorer where you downloaded the new .UF2 release and drag this .UF2 file to the RPi Pico file explorer. The code will be automatically installed and the file explorer should close. This implies successful installation of new software to the PicoMagnetometer detector.
- 14) Note, the new release will not impact the stored user configuration data held by the AT24C32 EEPROM – unless explicitly informed in the release notes associated with the new .UF2.
- 15) Replace the lower aluminium case of the PicoMagnetometer detector. Replace the x2 front panel lower case screws and the x2 rear panel lower aluminium case screws.
- 16) Upgrade complete – good to go.

Error codes and meaning.

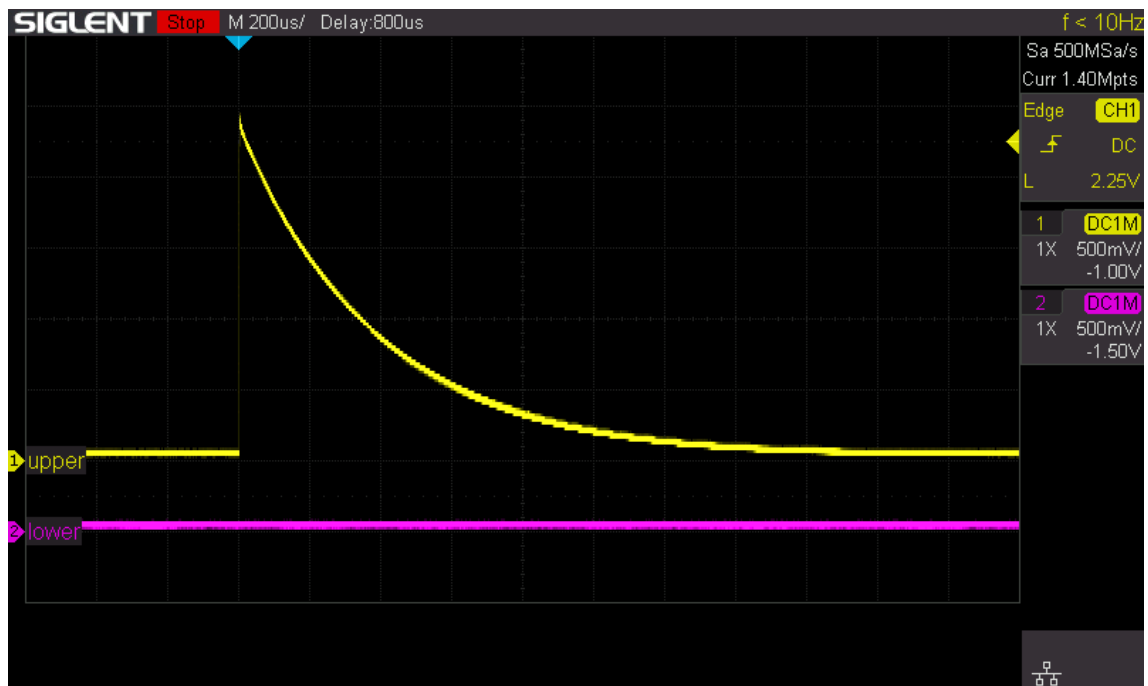
The PicoMagnetometer has a series of error codes that are displayed on the OLED display when there is a fault condition. Below is a table of error codes, cause and remedy.

Error code	Name	Cause	Remedy
ERR:00	ERROR_NONE	N/A	No action necessary.
ERR:01	ERROR_DS3231_READ	DS3231 read error – unable to read the external RTC.	Replace 2032 button cell and reset time.
ERR:02	ERROR_QUEUE_OVERFLOW	Internal debug message. Memory structure used to buffer up the events coming from the high speed event capture routine has overflowed.	Should never occur. Reboot if does.
ERR:03	ERROR_SD_CARD	SD Card interface failed to initialise.	Try different SD card. Reboot.
ERR:04	ERROR_SD_MOUNT	Failed to mount file system on SD card.	Check SD card file format.
ERR:05	ERROR_SD_FILE	Unable to create or access file on SD card.	Check SD card contents.
ERR:06	ERROR_SD_SPACE	Free space on SD card has dropped below 5.0 MB.	Free up space on SD card or replace with new one.
ERR:07	ERROR_RING_BUFF	Internal debug message indicating a ring buffer overflow.	Should never occur. Reboot if does.
ERR:08	ERROR_NO_CONFIG	Not used.	Should never occur.

Error code	Name	Cause	Remedy
ERR:09	ERROR_BAD_CONFIG	Invalid parameter/value detected in user configuration.	Check configuration content and fix.
ERR:10	ERROR_BAD_PRESSURE	Internal debug message. Invalid temperature and or pressure value detected.	Should never occur. Reboot if does.
ERR:21	ERROR_DEBUG_EVENTS_STOPPED	Internal debug message. Event capture has stopped.	Should never occur. Reboot if does.

Typical Oscilloscope Traces

- 1) Using the three sma connectors from the rear panel of the PicoMagnetometer detector you have access to the raw upper and lower event pulses, range in value from 0V to 3V3 (**threshold_t** and **threshold_b** set at 0.161V), that are read by the RPI Pico ADC, and to the coincidence pulse (0V to 3V3) generated by the RPI Pico when it identifies a coincidence event – this is a modulated pulse with direct connection to the coincidence LED.
- 2) The figure below shows a typical pulse trace for the upper SiPM signal following an event. Details of scope settings are included in the figure.



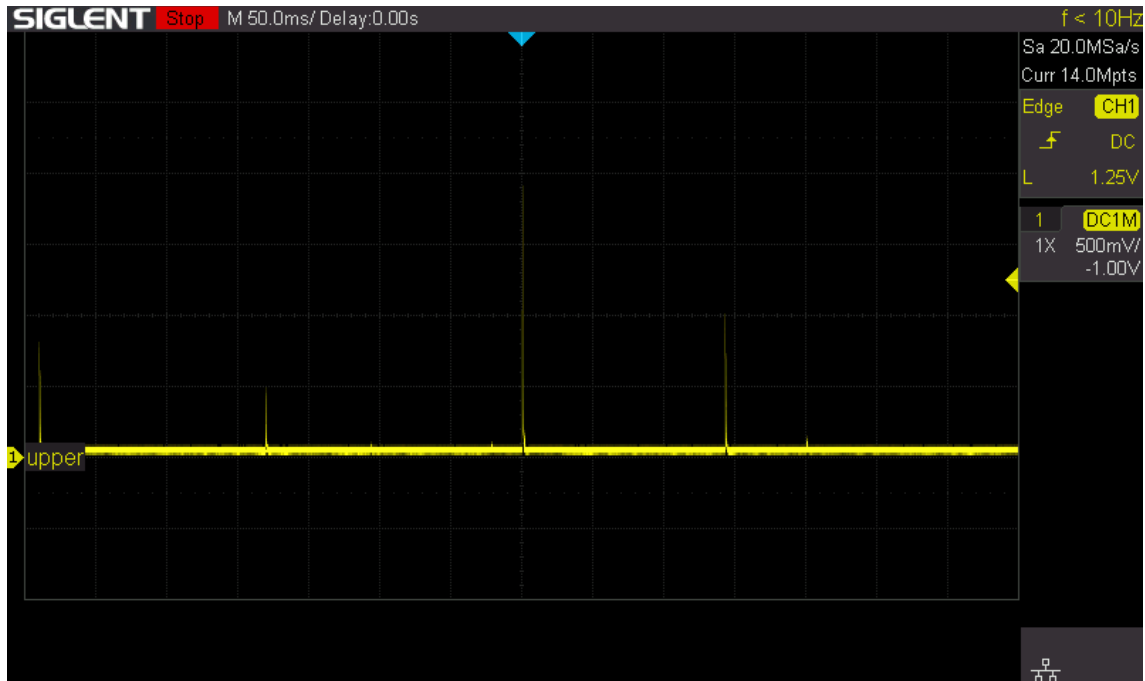
- 3) It can be seen in the figure above that the event pulse is approximately $8 \times 200\mu\text{s}$ length – 1.6s.

- 4) The figure below shows a typical pulse trace for the lower SiPM signal following an event. Again details of the scope settings are included in the figure.



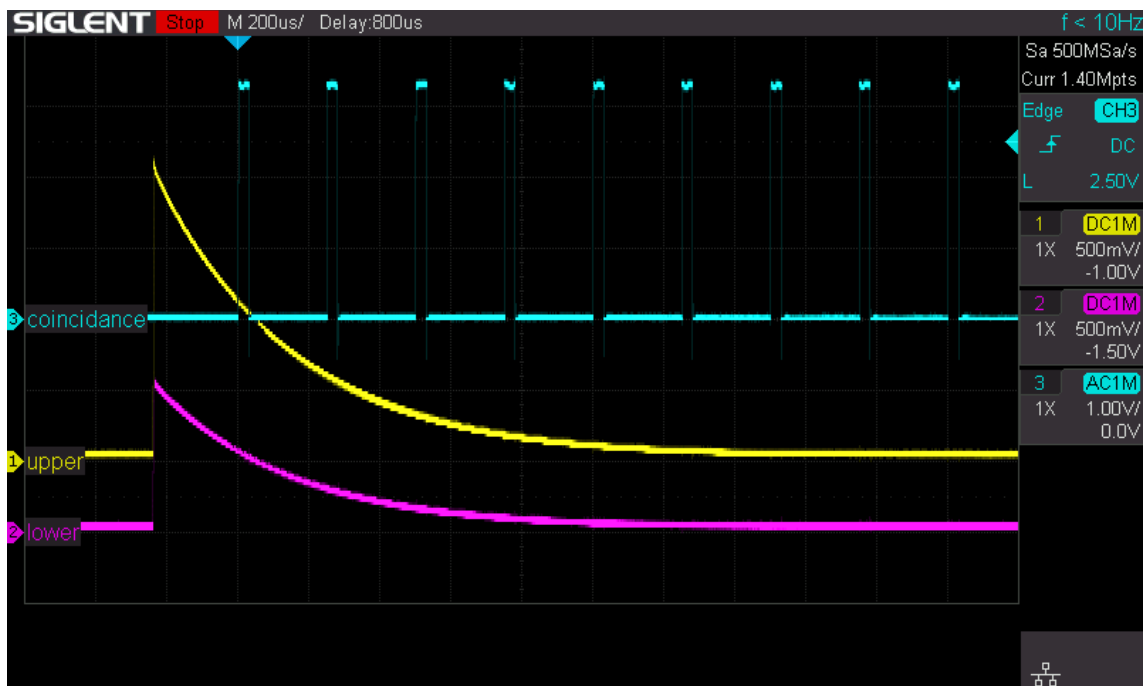
- 5) Again, it can be seen in the figure above that the event pulse is approximately 8 x 200 μ s length – 1.6s.

- 6) The following scope trace shows how the use of **threshold_t** and **threshold_b** is used to discriminate the recording of “noise” events recorded by the SiPMs.

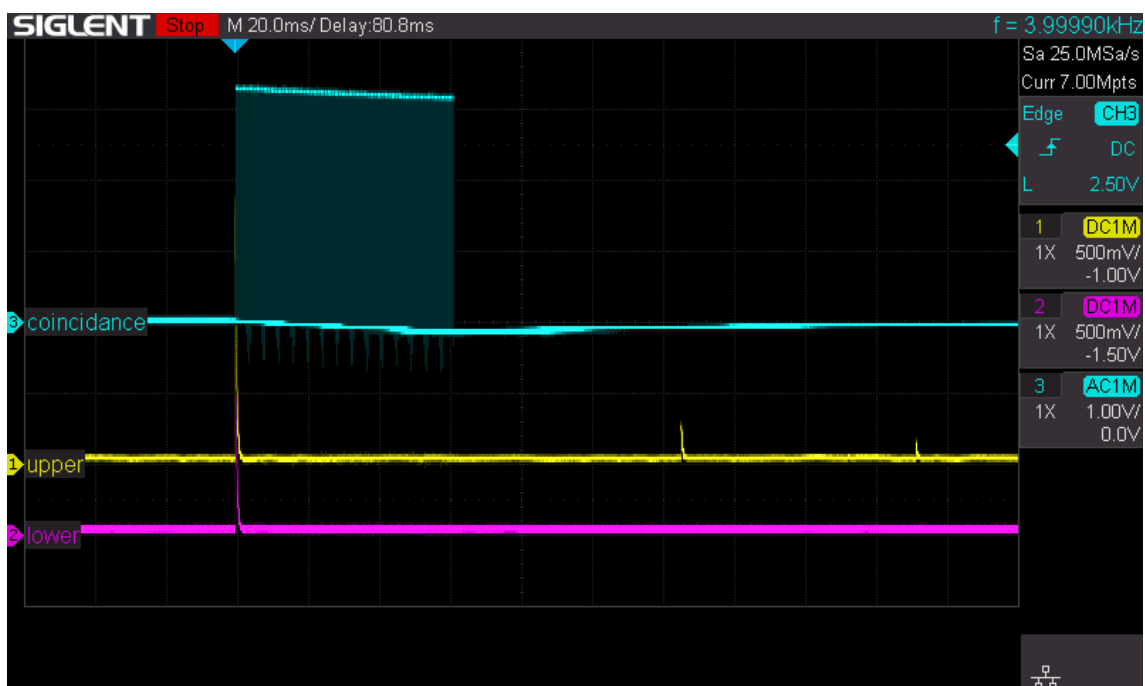


- 7) The trigger threshold of the scope traces above was set at 1.25V and it can be seen that the first two pulses on the left of the figure did not trigger the oscilloscope, these would not be recorded as events as they are below the threshold. It was the central pulse that triggered the oscilloscope.
- 8) It can also be seen that the amplitude of the two pulses on the right of the figure would equally have not triggered the oscilloscope, again, these would not be recorded as event as then are below the trigger threshold.

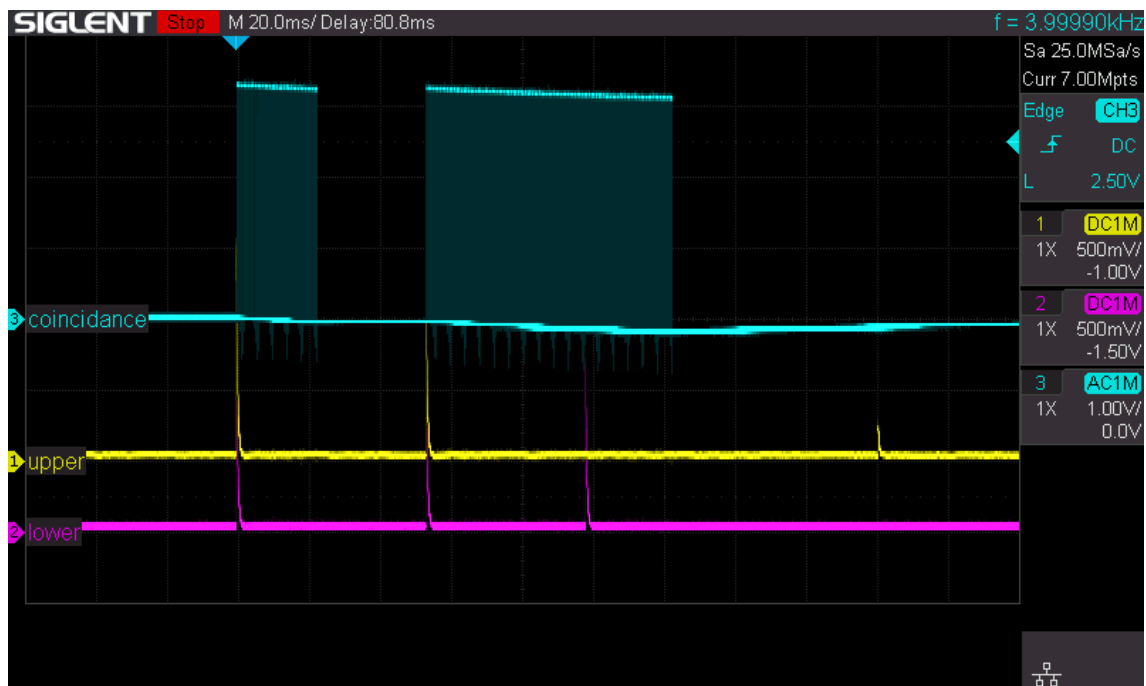
- 9) The scope trace below illustrates a coincidence event – note that there is about $200\mu\text{s}$ between the upper/lower pulse event registration and the output of the coincidence sma going high. This is a pulsed high/low output (PWM) as it also drives the front coincidence LED.



- 10) The scope trace below shows that the coincidence sma output is pulsed high/low for approximately 60ms (from 0V to 3V3) following a coincidence event.



11) The scope trace below records two coincidence events within a period of about 50ms.



12) There is a third event recorded by the lower SiPM, at about 100ms from the first event, but no event on the upper SiPM recorded at the same time – hence not a coincidence event.

13) There is also a third event shown of the upper SiPM trace, at about 180ms from the first event, but this is below the threshold value and would not be recorded by the PicoMagnetometer detector as an event as it is below the threshold level.

Contacts

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Registered Charity in England and Wales No. 1123866.

E-mail: Magnetometer@ukraa.com

Website: ukraa.com

BAA Radio Astronomy Group

Website: BAA Radio Astronomy

Magnetometer forum

Website: UKRAAmagnetometer@groups.io

Appendix 1 – PicoMagnetometer Specifications

Basic information	Size: 104mm x 85mm x 60mm Mass: 350g
Power requirements	Type: Via micro USB to RPi Pico board Voltage: 5V DC Current peak: 150mA Current nominal: 70mA
Fluxgate sensor	Type: FLC-100 Emission λ_{max}: 425nm Size: 50mm x 50mm x 10mm (nominal) Rise time: 0.9ns Decay time: 2.1ns
Processor	Model: Raspberry Pi Pico2 Core: RP2350 microcontroller Speed: Up to 150MHz SRAM: 256KB Flash memory: 4MB

Display	Type: OLED Size: 1.5" Pixels: 128×128 Drive: sh1107 Interface: I2C
Environment monitoring	Type: TMP36 temperature sensor Interface: I2C Temperature range: −40°C to +85°C
Real Time clock	Type: DS3231 I2C–integrated RTC Interface: I2C Accuracy: About 30 minutes per year
EEPROM	Type: AT24C32 2–wire Serial EEPROM Interface: I2C Capacity: 4kB @128 pages of 32 bytes
SD card slot	Type: microSD card size Interface: SPI Capacity: Tested to 64GB
Maximum count rate	Top detector: 30cps Bottom detector: 30cps

Appendix 2 – Regulatory Compliance

RoHS

In Great Britain and Northern Ireland, the Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012 (as amended) are the underpinning legislation. The Restriction of Hazardous Substances Directive or RoHS regulations restrict of the use of the following ten hazardous substances in electronic and electrical equipment products falling within the Directive:

- Lead
- Mercury
- Cadmium
- Hexavalent chromium
- Polybrominated biphenyls
- Polybrominated diphenyl ethers
- Bis(2-ethylhexyl) phthalate (DEHP)
- Butyl benzyl phthalate (BBP)
- Dibutyl phthalate (DBP)
- Diisobutyl phthalate (DIBP)

UKRAA confirms that the suppliers of the components and materials used in the UKRAA PicoMagnetometer detector have stated that such components and materials are RoHS compliant and that reasonable steps have been taken to confirm these statements.

WEEE

Electrical and electronic equipment (EEE) is regulated to reduce the amount of waste electrical and electronic equipment (WEEE) incinerated or sent to landfill sites. Reduction is achieved through various measures which encourage the recovery, reuse and recycling of products and components. The Waste Electrical and Electronic Equipment Regulations 2013 (as amended) is the underpinning UK legislation. (gov.uk website)

If a customer purchases a new product from UKRAA which falls within the WEEE Directive to replace an existing one (of similar function to the one that has been sold) and intends to dispose of the existing one, then the customer can request that we take back the existing product and recycle it. Any customer wishing to take advantage of this facility should contact us.

You may also use alternative local WEEE recycling provision. Please do not place WEE in the general waste.

Revision History

Revision	Date	Author	Status
Draft A	2026-08-05	R Knott	Internal draft for peer review

Outstanding work

LOTS!!!



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